

Decomposable Octahedrality Is Dual to the Weak-Star Symmetric Strong Diameter-Two Property

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Abstract

Ostrak proved that if X^* has the weak-star symmetric strong diameter-two property, then X is decomposably octahedral, and asked whether the converse holds for every Banach space. We prove the converse. The simultaneous dual-ball constraints are encoded as the unit ball of the norm

$$(f_1, \dots, f_n, g) \longmapsto \max_i \{ \|f_i + g\|, \|f_i - g\| \}.$$

Hahn–Banach expresses its support function by decompositions in X^{**} . The principle of local reflexivity transfers each finite decomposition to X , where decomposable octahedrality supplies exactly the required lower bound. A finite-dimensional separation argument then produces one large functional g that works simultaneously in all prescribed weak-star slices. Consequently, for every real Banach space X , X is decomposably octahedral if and only if X^* has the weak-star symmetric strong diameter-two property.

1 The question and the result

All Banach spaces are real. A weak-star slice of B_{X^*} is

$$S(B_{X^*}, x, \alpha) = \{f \in B_{X^*} : f(x) > 1 - \alpha\}, \quad x \in S_X, \quad \alpha > 0.$$

The dual X^* has the *weak-star symmetric strong diameter-two property* (w^* -SSD2P) if, for every finite family of weak-star slices $S_1, \dots, S_n$ and every $\varepsilon > 0$, there are $f_i \in S_i$ and $g \in B_{X^*}$ such that

$$f_i + g \in S_i, \quad f_i - g \in S_i \quad (1 \leq i \leq n), \quad \|g\| > 1 - \varepsilon.$$

Following Ostrak [1], X is *decomposably octahedral* (DOH) if, for every finite $E \subset S_X$ and every $\rho > 0$, there is $y \in S_X$ such that, whenever

$$y = \sum_{i=1}^m y_i, \quad x_i \in E, \quad a_i, b_i \geq 0,$$

one has

$$\sum_{i=1}^m (\|a_i x_i + y_i\| + \|b_i x_i - y_i\|) \geq (1 - \rho) \left(\sum_{i=1}^m (a_i + b_i) + 2 \right). \quad (1)$$

Ostrak proved that w^* -SSD2P of X^* implies DOH of X and asked whether the converse holds [1, Proposition 2.1 and p. 7]. The question was still recorded as open in 2024 [2, Section 4, paragraph preceding Question 4.2].

Theorem 1 (Full converse). *Let X be a real Banach space. If X is decomposably octahedral, then X^* has the weak-star symmetric strong diameter-two property.*

Combined with Ostrak's proposition, this gives the exact duality.

Corollary 2. *For every real Banach space X , the following are equivalent:*

- (i) X is decomposably octahedral;
- (ii) X^* has the weak-star symmetric strong diameter-two property.

2 Proof intuition

Fix slice directions $x_1, \dots, x_n$. We seek functionals f_i and a common g for which both $f_i + g$ and $f_i - g$ lie in the dual unit ball and are almost norming at x_i . Add one more desired coordinate, $g(y)$, where y is supplied by DOH. If the resulting compact convex set of $2n + 1$ coordinates missed the upper orthant near $(1, \dots, 1)$, a nonnegative linear functional would separate them.

It therefore suffices to lower-bound every nonnegative support functional. The constraints $\|f_i \pm g\| \leq 1$ define the unit ball of an equivalent norm on $(X^*)^{n+1}$. Hahn–Banach computes its dual norm as a minimum of

$$\sum_i (\|a_i Jx_i + r_i\| + \|b_i Jx_i - r_i\|), \quad 2 \sum_i r_i = cJy,$$

over $r_i \in X^{**}$. Formula (1) is precisely the needed lower bound when the r_i belong to X . Local reflexivity sends any finite family in X^{**} almost isometrically into X , fixes the displayed vectors from X , and preserves the sum. Letting the local-reflexivity distortion tend to one proves the same bound in X^{**} . This closes the duality gap and makes separation impossible.

3 The support-function formula

Write $J : X \rightarrow X^{**}$ for the canonical embedding.

Lemma 3 (Hahn–Banach support formula). *Fix $x_1, \dots, x_n, y \in X$ and define*

$$\mathcal{C} = \left\{ (f_1, \dots, f_n, g) \in (X^*)^{n+1} : \|f_i + g\| \leq 1, \|f_i - g\| \leq 1 \ (1 \leq i \leq n) \right\}.$$

For $a_i, b_i, c \geq 0$, let

$$H(a, b, c) = \sup_{(f, g) \in \mathcal{C}} \left[\sum_{i=1}^n a_i (f_i + g)(x_i) + \sum_{i=1}^n b_i (f_i - g)(x_i) + cg(y) \right].$$

Then

$$H(a, b, c) = \min_{\substack{r_1, \dots, r_n \in X^{**} \\ 2 \sum_i r_i = cJy}} \sum_{i=1}^n (\|a_i Jx_i + r_i\| + \|b_i Jx_i - r_i\|). \quad (2)$$

Proof. On $Z = (X^*)^{n+1}$ define

$$N(f_1, \dots, f_n, g) = \max_{1 \leq i \leq n} \{\|f_i + g\|, \|f_i - g\|\}.$$

The map

$$A : Z \longrightarrow W = (X^*)^{2n}, \quad A(f_1, \dots, f_n, g) = (f_i + g, f_i - g)_{i=1}^n,$$

is an isometry when W has its maximum norm; its range is closed because f_i and g are recovered by taking half-sums and half-differences. Thus $\mathcal{C}$ is the N -unit ball.

The displayed functional defining $H(a, b, c)$ is an element of Z^* . Since the N -unit ball is symmetric, $H(a, b, c)$ is its dual norm. By Hahn–Banach, this norm is the least norm of an extension from $A(Z)$ to W . Since W^* is the ℓ_1 -sum of $2n$ copies of X^{**} , such an extension is represented by $p_i, q_i \in X^{**}$ and has norm $\sum_i (\|p_i\| + \|q_i\|)$. The restriction agrees with the displayed functional exactly when

$$p_i + q_i = (a_i + b_i)Jx_i, \tag{3}$$

$$\sum_{i=1}^n (p_i - q_i) = \sum_{i=1}^n (a_i - b_i)Jx_i + cJy. \tag{4}$$

Put $r_i = p_i - a_i Jx_i$. Equations (3)–(4) become

$$p_i = a_i Jx_i + r_i, \quad q_i = b_i Jx_i - r_i, \quad 2 \sum_i r_i = cJy.$$

Conversely, every such family (r_i) gives an extension. Hahn–Banach provides an extension of exactly the dual norm, so the minimum is attained and (2) follows. $\square$

4 DOH controls the bidual decompositions

Lemma 4 (Local-reflexivity transfer). *Assume that X is DOH. Fix a finite set $E = \{x_1, \dots, x_n\} \subset S_X$ and $\rho > 0$, and choose $y \in S_X$ as in (1). Then, for every $a_i, b_i, c \geq 0$,*

$$H(a, b, c) \geq (1 - \rho) \left(\sum_{i=1}^n (a_i + b_i) + c \right), \tag{5}$$

where H is defined in Lemma 3.

Proof. By Lemma 3, it is enough to bound the objective in (2) for an arbitrary admissible family (r_i) .

If $c = 0$, then $\sum_i r_i = 0$, although even this equality is unnecessary: the triangle inequality gives, term by term,

$$\|a_i Jx_i + r_i\| + \|b_i Jx_i - r_i\| \geq (a_i + b_i)\|Jx_i\| = a_i + b_i.$$

Suppose that $c > 0$. Let $F \subset X^{**}$ be the finite-dimensional span of $r_1, \dots, r_n, Jx_1, \dots, Jx_n, Jy$. The principle of local reflexivity [3] says that, for every $\tau > 0$, there is a linear map $T_\tau : F \rightarrow X$ such that

$$T_\tau(Jx) = x \quad (Jx \in F \cap JX), \quad \|T_\tau\| \leq 1 + \tau.$$

Set $y_i = (2/c)T_\tau r_i$. Since $2 \sum_i r_i = cJy$ and $T_\tau(Jy) = y$, we have $\sum_i y_i = y$. Applying (1) gives

$$\begin{aligned} & \sum_i (\|a_i x_i + T_\tau r_i\| + \|b_i x_i - T_\tau r_i\|) \\ &= \frac{c}{2} \sum_i \left(\left\| \frac{2a_i}{c} x_i + y_i \right\| + \left\| \frac{2b_i}{c} x_i - y_i \right\| \right) \\ &\geq (1 - \rho) \left(\sum_i (a_i + b_i) + c \right). \end{aligned}$$

On the other hand, because T_τ fixes each Jx_i ,

$$\begin{aligned} & \sum_i (\|a_i Jx_i + r_i\| + \|b_i Jx_i - r_i\|) \\ & \geq \frac{1}{1+\tau} \sum_i (\|a_i x_i + T_\tau r_i\| + \|b_i x_i - T_\tau r_i\|). \end{aligned}$$

Letting $\tau \downarrow 0$ proves the desired lower bound for every admissible (r_i) . Formula (2) now yields (5). $\square$

5 Separation and completion of the proof

Proof of Theorem 1. Let

$$S_i = S(B_{X^*}, x_i, \alpha_i), \quad 1 \leq i \leq n,$$

be weak-star slices, and fix $\varepsilon > 0$. Choose

$$0 < \rho < \min\{\varepsilon, \alpha_1, \dots, \alpha_n\},$$

and let $y \in S_X$ be supplied by DOH for $E = \{x_1, \dots, x_n\}$ and the parameter ρ .

For the set $\mathcal{C}$ in Lemma 3, consider the compact convex subset $K \subset \mathbb{R}^{2n+1}$ consisting of the vectors

$$((f_i + g)(x_i), (f_i - g)(x_i))_{i=1}^n, g(y), \quad (f_1, \dots, f_n, g) \in \mathcal{C}.$$

Indeed, $\mathcal{C}$ is weak-star closed and lies in the product of $n+1$ dual unit balls: both f_i and g are half-sums or half-differences of $f_i + g$ and $f_i - g$. Banach–Alaoglu therefore gives the asserted compactness.

We claim that

$$K \cap [1 - \rho, \infty)^{2n+1} \neq \emptyset. \quad (6)$$

Otherwise, finite-dimensional strong separation of the compact set K from the closed upper orthant gives coefficients $a_i, b_i, c \geq 0$, not all zero, such that

$$\begin{aligned} \sup_{(f,g) \in \mathcal{C}} \left[\sum_i a_i (f_i + g)(x_i) + \sum_i b_i (f_i - g)(x_i) + cg(y) \right] \\ < (1 - \rho) \left(\sum_i (a_i + b_i) + c \right). \end{aligned}$$

The coefficients are nonnegative because a linear functional bounded below on the upper orthant must have nonnegative coordinates. The strict inequality contradicts Lemma 4. This proves (6).

Choose $(f_1, \dots, f_n, g) \in \mathcal{C}$ producing a point in the intersection. Then

$$\|f_i \pm g\| \leq 1, \quad (f_i \pm g)(x_i) \geq 1 - \rho > 1 - \alpha_i.$$

Thus $f_i + g, f_i - g \in S_i$. Their midpoint f_i also belongs to the convex slice S_i . Finally,

$$\|g\| \geq g(y) \geq 1 - \rho > 1 - \varepsilon.$$

These are exactly the requirements of the w^* -SSD2P. $\square$

Proof of Corollary 2. The implication (ii) $\Rightarrow$ (i) is [1, Proposition 2.1]; the reverse implication is Theorem 1. $\square$

6 Audit note

The point at which a naive duality proof can appear to fail is important. Hahn–Banach naturally produces $r_i \in X^{**}$, whereas the definition of DOH quantifies only over decompositions in X . The finite-dimensional span of a single candidate decomposition is precisely the setting of local reflexivity. Because the sum of the r_i is the canonical image of the fixed vector $cy/2$, a local-reflexivity map both preserves that sum and fixes every x_i . No approximation of infinitely many vectors and no compactness passage is needed.

References

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